

Day 18 - Lecture 18.14: Advanced Customizations on Scatter Plots

1. Overview & Semantic Encoding

Scatter plots can convey **up to 4 dimensions of data** simultaneously on a 2D plane:

1. **X-position**: Dimension 1 (e.g. Age)
 2. **Y-position**: Dimension 2 (e.g. Blood Pressure)
 3. **Point Area (s)**: Dimension 3 (e.g. Magnitude / Severity)
 4. **Color Mapping (c + cmap)**: Dimension 4 (e.g. Risk Level)
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2. Dynamic Colors & Continuous Colormaps

1. Conditional Categorical Colors (List Comprehension):

```
colors = ["green" if bp < 135 else "red" for bp in blood_pressure]
plt.scatter(age, blood_pressure, color=colors)
```

2. Continuous Colormap (cmap) with Colorbar:

```
plt.scatter(age, blood_pressure, c=blood_pressure, cmap="OrRd", s=blood_pressure)
plt.colorbar(label="Blood Pressure Severity")
```

- Available colormaps: 'viridis', 'plasma', 'inferno', 'OrRd', 'coolwarm'.
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3. Code Implementation

```
import matplotlib.pyplot as plt

age = [22, 25, 30, 35, 40, 45, 50, 55, 60, 65]
blood_pressure = [110, 115, 120, 122, 125, 130, 135, 123, 145, 150]

# 1. Discrete Conditional Colors + Dynamic Size
colors = ["seagreen" if bp < 135 else "crimson" for bp in blood_pressure]
sizes = [bp * 1.5 for bp in blood_pressure]

plt.figure(figsize=(9, 5))
plt.scatter(age, blood_pressure, color=colors, s=sizes, marker='^', alpha=0.7, ec='black')
plt.title("Conditional Styling: Green = Normal (<135), Red = Elevated (>=135)", fontweight="bold")
plt.xlabel("Age")
plt.ylabel("Blood Pressure")
plt.grid(True, linestyle=":", alpha=0.6)
plt.show()

# 2. Continuous Colormap with Colorbar
plt.figure(figsize=(9, 5))
scatter = plt.scatter(age, blood_pressure, c=blood_pressure, cmap="OrRd", s=120, ec='black')
cbar = plt.colorbar(scatter)
cbar.set_label("Blood Pressure (mmHg)", fontsize=11)

plt.title("Color by Value using Colormap ('OrRd')", fontsize=14, fontweight="bold")
plt.xlabel("Age")
plt.ylabel("Blood Pressure")
plt.grid(True, linestyle="--", alpha=0.5)
plt.show()
```

4. Key Takeaways

- `c=` accepts an array of numerical values mapped through `cmap`, whereas `color=` takes static color strings or an array of literal color names.
- Always include `plt.colorbar()` when using `cmap` so viewers understand what the gradient represents.